

Functional Annotation Report: ttg2E (PP_0962, UniProt Q88P90)

Pseudomonas putida KT2440

1. Summary / Answer

ttg2E (PP_0962; Q88P90) encodes the MlaB ortholog — a small (100-residue) cytoplasmic STAS-domain protein that is the auxiliary regulatory/assembly subunit of the inner-membrane ABC transporter MlaFEDB (the "Ttg2" system) in *Pseudomonas putida* KT2440. It is not itself an enzyme or a transporter; instead it binds the MlaF ATPase on the cytoplasmic face of the inner membrane, where it is required for both the correct assembly and the ATP-hydrolytic activity of the transporter. The MlaFEDB complex powers **retrograde transport of mislocalized glycerophospholipids from the outer membrane (OM) back to the inner membrane (IM)**, thereby maintaining OM lipid asymmetry. This membrane-barrier function is the molecular basis for the historically observed "toluene-tolerance" phenotype and for low-level intrinsic resistance to lipid-permeant antibiotics.

2. Gene / Protein Identity Verification (mandatory)

The verification requested in the task was performed and **passed**:

- **Gene symbol match:** The gene is named `ttg2E` and is annotated "Toluene-tolerance protein." The "ttg" (toluene-tolerance genes) label originates from a transposon screen in *P. putida* that first identified `ttg` loci required to survive solvent exposure (Kim et al. 1998, [P 9658016](#)), in which Ttg1/Ttg2 were identified as ABC-transporter homologs. The label is a **phenotype-based historical name**, not a description of the biochemical activity.
- **Organism confirmed:** *Pseudomonas putida* strain KT2440 (taxid 160488); locus tag PP_0962.
- **Domain/family match:** UniProt/InterPro assign a single **STAS domain** (Pfam PF13466 `STAS_2`; InterPro IPR002645 / IPR036513; SUPFAM SSF52091 SpoIIaa-like; Gene3D

3.30.750.24). A ~100-aa protein consisting solely of a STAS domain is the diagnostic architecture of **MlaB**.

- **Operon context (decisive):** PP_0962 is the last gene of the operon **PP_0958–PP_0962**, which UniProt annotates as:

Locus	Gene	Len (aa)	Pfam	Mla component
PP_0958	mlaF	269	PF00005 (ABC ATPase)	MlaF (ATPase / NBD)
PP_0959	mlaE	266	PF02405	MlaE (permease / TMD)
PP_0960	mlaD	161	PF02470 (MCE)	MlaD (substrate-binding)
PP_0961	ttg2D	215	PF05494	MlaC (periplasmic shuttle)
PP_0962	ttg2E	100	PF13466 (STAS)	MlaB (regulatory subunit)

This gene order (F-E-D-C-B) is identical to the canonical *E. coli* **mlaFEDCB** operon, unambiguously assigning **ttg2E** = **MlaB**. The "Ttg2 = Mla" equivalence is independently confirmed by the demonstration that *P. aeruginosa* Ttg2D is the periplasmic phospholipid-binding (MlaC-type) component of this ABC system (Yero et al. 2021,).

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No conflicting-gene ambiguity was encountered. All lines of evidence converge on a single, well-characterized system.

3. Primary Function

3.1 Molecular role — a STAS-domain adapter/regulator, not an enzyme

MlaB is a cytoplasmic **STAS (Sulphate Transporter and Anti-Sigma-factor antagonist) domain** protein. Direct analysis of the *P. putida* sequence (Q88P90) shows a 100-residue protein consisting of a **single STAS domain (residues 13–100)** with **no signal peptide and no transmembrane segment** (longest hydrophobic stretch = 6 residues, well below the ~15–20 needed for a membrane helix) — i.e. a small, soluble, cytoplasmic protein. In the assembled transporter it clamps onto the nucleotide-binding (ATPase) domain MlaF (typically two MlaB copies per MlaFEDB complex). Functional dissection of the homologous *E. coli* complex showed that the transporter is built from canonical components MlaF and MlaE plus **auxiliary proteins MlaD and MlaB**, and that **"MlaB plays critical roles in both the assembly and**

activity of the transporter" (Thong et al. 2016,).

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A subsequent explosion of **cryo-EM structures** of MlaFEDB (apo, phospholipid-, ADP- and AMP-PNP-bound; 3.3–4.1 Å) resolves MlaB as an integral subunit of the complex, and mutagenesis mapped residues that "regulate the activity and structural stability of the MlaFEDB complex" (Tang et al. 2021,

[P 33199922](#)

reviewed in Ekiert et al. 2022,).

[P 35981415](#)

Thus the precise role of ttg2E is:

- **Structural/assembly:** stabilizes the MlaFEDB complex; without MlaB the complex fails to assemble properly.
- **Regulatory:** modulates/couples ATP hydrolysis by MlaF that energizes lipid transport.

It has **no substrate specificity of its own** (it is not a catalytic or transport subunit); its "substrate" is effectively its protein partner MlaF.

3.2 Pathway — Maintenance of Lipid Asymmetry (Mla)

The MlaFEDB transporter is the inner-membrane engine of the cell-envelope-spanning **Mla pathway**, which comprises three assemblies: the OM lipoprotein complex MlaA–OmpC/E, the periplasmic phospholipid shuttle MlaC (= Ttg2D here), and the IM ABC transporter **MlaFEDB** (containing MlaB) (Abellon-Ruiz 2023,

[P 36459067](#)

The pathway removes glycerophospholipids that become mislocalized in the OM outer leaflet and returns them to the inner membrane — i.e. **retrograde (OM→IM) phospholipid transport** — using energy from ATP hydrolysis to maintain OM lipid asymmetry (Thong et al. 2016,

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MlaB's function is therefore executed *within* this pathway by enabling the ATPase that drives the shuttle.

3.3 Physiological output — membrane barrier integrity

By preserving OM lipid asymmetry (LPS outer leaflet / phospholipid inner leaflet), the Ttg2/Mla system keeps the OM an effective permeability barrier. This explains the associated phenotypes: - **Organic-solvent (toluene) tolerance** in *P. putida*: `ttg2` disruption yields toluene-hypersensitive mutants, and membrane integrity/repair — not solvent efflux per se — was identified as the key protective mechanism (Kim et al. 1998, [P 9658016](#)).

- **Intrinsic antibiotic resistance**: in *P. aeruginosa* the Ttg2 system contributes to low-level resistance against antibiotics that permeate via a lipid-mediated pathway (Yero et al. 2021, [P 33837253](#)).

4. Localization

- **Sub-cellular**: MlaB/ttg2E is a **cytoplasmic** protein peripherally associated with the **cytoplasmic face of the inner membrane**, as part of the MlaFEDB complex (the ATPase MlaF and its MlaB partner face the cytoplasm; MlaE spans the IM; MlaD projects into the periplasm)

([P 36459067](#));

[P 27529189](#));

visualized in cryo-EM structures

[P 33199922](#)).

This is directly supported by the sequence: a soluble single STAS domain (res 13–100) with no signal peptide and no transmembrane helix in the 100-aa protein.

5. Evidence Summary

Claim	Evidence type	Source
ttg2E = MlaB (STAS subunit)	Genomic/operon mapping + domain annotation (UniProt/Pfam/InterPro)	This analysis; UniProt Q88P90
Ttg2 system = Mla lipid- asymmetry ABC transporter in <i>Pseudomonas</i>	Crystal structure + biochemistry of Ttg2D=MlaC	Yero et al. 2021, P 33837253
MlaB required for assembly + ATPase activity of MlaFEDB	Genetic + biochemical reconstitution (E. coli homolog)	Thong et al. 2016, P 27529189
MlaB is an integral subunit of MlaFEDB; regulates complex activity/stability	Cryo-EM (3.3–4.1 Å) + mutagenesis (E. coli homolog)	Tang et al. 2021, P 33199922 ; Ekiert et al. 2022, P 35981415
ttg2E is soluble/cytoplasmic (single STAS domain, no TM/signal)	Sequence/bioinformatic analysis of Q88P90	This analysis (UniProt Q88P90)
Pathway = retrograde OM→IM phospholipid transport maintaining OM asymmetry	Review of structural/ biochemical literature	Abellon-Ruiz 2023, P 36459067 ; Thong 2016
ttg2 needed for toluene tolerance via membrane integrity	Tn5 mutagenesis in <i>P. putida</i>	Kim et al. 1998, P 9658016
Ttg2/Mla → intrinsic antibiotic resistance	Gene knockouts in <i>P. aeruginosa</i>	Yero et al. 2021, P 33837253
ttg2/Mla is distinct from RND solvent-efflux pumps (TtgABC etc.)	Authoritative review of solvent-tolerance mechanisms	Ramos et al. 2002, P 12142492

Evidence spans **direct experiment** (genetic reconstitution, crystallography, knockouts) in closely related orthologs and **bioinformatic inference** (operon synteny, conserved domain architecture) for the KT2440 protein specifically. No experimental study on the KT2440 MlaB protein itself was found; the annotation is by strong orthology.

6. Supported and Refuted Hypotheses

- **Supported:** ttg2E encodes MlaB, a cytoplasmic STAS-domain auxiliary subunit of the inner-membrane MlaFEDB (Ttg2) ABC transporter that maintains OM lipid asymmetry.
- **Refuted / clarified:** The literal annotation "toluene-tolerance protein" does **not** mean ttg2E is a solvent-efflux transporter or a solvent-sensing/detoxifying enzyme. The solvent-tolerance phenotype is an indirect consequence of maintaining OM barrier integrity. An authoritative review of Gram-negative solvent tolerance (Ramos et al. 2002,

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) frames the phenomenon in terms of (a) membrane phospholipid/fatty-acid adaptation, (b) vesiculation, and (c) **RND-family** active efflux pumps that export solvents. ttg2E/MlaB contributes to the phospholipid-homeostasis arm (category a), whereas the well-known *P. putida* solvent-efflux pumps are the mechanistically **separate RND systems TtgABC / TtgDEF / TtgGHI** — a different protein superfamily that must not be conflated with the ttg2/Mla ABC transporter, despite the shared "ttg" (toluene-tolerance genes) label.

7. Limitations and Future Directions

- Functional data are from orthologs (*E. coli*, *P. aeruginosa*); the KT2440 MlaB has not been individually characterized experimentally, though orthology is unambiguous.
- The direction of net lipid transport (retrograde vs. anterograde) remains debated in the field, though current biochemistry favors retrograde; this affects mechanistic detail but not MlaB's role as an MlaF-regulatory subunit.
- Useful follow-up: co-purify/crystallize the *P. putida* MlaFEDB(ttg2E) complex; measure ATPase activity ± MlaB; test a clean PP_0962 deletion for OM permeability, vesiculation, solvent and antibiotic sensitivity.